ClinicalTrials.gov Viz Agent - Database Schema (Postgres)

PK = primary key, FK = foreign key, UQ = unique (partial), dashed = logical reference (no FK)

registry.term_sets (a term's trials, Q22)

term_key	text	PK
param, expression	text	
size	integer	
data_timestamp	timestamptz	
resolved_at, checked_at	timestamptz	

1 term : N trials

registry.term_members

term_key	text	PK, FK
nct_id	text	PK, idx

registry.sync_runs (loads and updates)

run_id	integer	PK
kind	text (loadlupdate)	
status	text (runninglsucceededlfailed)	idx
data_timestamp	timestamptz	
since	date (nullable)	
upserted, removed	integer	
error	text (nullable)	
started_at, finished_at	timestamptz	idx

registry.trials (the mirror, Q22)

nct_id	text	PK
record	jsonb (the API's study)	
start_year	integer (nullable)	idx
last_update_post_date	date (nullable)	
synced_at	timestamptz	

plan_cache (L2, standalone)

request_key	char(64)	PK
planner_version	text	
plan	jsonb	
planner_trace	jsonb	
created_at	timestamptz	
last_used_at	timestamptz	
hit_count	integer	

data_versions

data_timestamp	timestamptz	PK
api_version	text	
first_seen_at	timestamptz	
last_checked_at	timestamptz	

verified against

verified against

api_searches (L1 cache, search level; not built)

search_key	char(64)	PK
endpoint	text	
params	jsonb	
kind	text (countlpageslstatslstudy)	
total_count	integer	
pages_expected	integer	
complete	boolean	
verified_data_ts	timestamptz	FK
fetched_at	timestamptz	
last_verified_at	timestamptz	
last_used_at	timestamptz	

1 search : N pages

api_pages (raw pages; not built)

search_key	char(64)	PK, FK
page_index	integer	PK
n_studies	integer	
body_gz	bytea (gzip raw bytes)	
raw_bytes	integer	
fetched_at	timestamptz	

results (L3, immutable rows)

result_id	uuid	PK
result_key	char(64)	idx
status	text (oklno_matching_trialsl...)	
request	jsonb	
plan	jsonb	
response	jsonb	
search_keys	char(64)[] (built as searches jsonb)	logical
verified_data_ts	timestamptz	FK
schema_version	text	
code_version	text	
created_at	timestamptz	
expires_at	timestamptz	idx

1 result : N data points

result_datums (one per data point)

result_id	uuid	PK, FK
datum_key	text	PK
label	text	
value	double precision	
citation_count	integer	
source	jsonb (nullable: API search of a sampled point)	
next_page_token	text (nullable)	
complete	boolean (default true)	

1 data point : N citations

result_citations (keyset-paginated)

result_id	uuid	PK, FK
datum_key	text	PK, FK
rank	integer	PK
nct_id	char(11)	
title	text	
evidence	jsonb [{field, excerpt}]	

jobs (background tasks)

job_id	uuid	PK
request_key	char(64)	UQ (active)
request	jsonb	
plan	jsonb	
status	text (pendinglrunninglsucceededlfailedlcancelled)	idx
progress	jsonb	
estimated_seconds	integer	
result_id	uuid (nullable)	FK
error	jsonb	
attempts / max_attempts	integer	
locked_by	text	
heartbeat_at	timestamptz	
trace_id	uuid	
client_ip	text (nullable)	
created_at / started_at / finished_at	timestamptz	
expires_at	timestamptz	

traces (observability; not built)

trace_id	uuid	PK
kind	text (requestljob)	
created_at	timestamptz	idx
request	jsonb	
http_status	integer	
outcome	text	
guard	jsonb	
stages	jsonb	
api_calls	jsonb	
result_id	uuid (nullable)	FK
job_id	uuid (nullable)	FK

llm_usage (one row per LLM call)

call_id	uuid	PK
trace_id	uuid (nullable)	FK
job_id	uuid (nullable)	FK
stage	text (guardlplannerlnarrator)	
model	text	idx
attempt	integer (not built)	
is_fallback	boolean	
status	text (oklerrorltimeout; not built)	
input_tokens	integer	
cached_input_tokens	integer	
output_tokens	integer	
reasoning_tokens	integer	
latency_ms	integer	
cost_usd	numeric(12,6)	
price_version	text	
created_at	timestamptz	idx